



Zoo $\leftrightarrow$ Lux Bridge: Multi-Chain Asset Transfer via Warp Messaging

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February 2025

Abstract

We present the cross-chain bridge protocol between Zoo L2 (Chain ID: 200200) and Lux Network (C-Chain, X-Chain, B-Chain). The bridge is built on the Lux Teleport architecture (LP-6332) and Warp messaging (ZIP-0022), and secured by MPC threshold signatures with a $\lceil 2n/3 \rceil$ supermajority requirement using the GG20 threshold ECDSA scheme. It supports bridging of ZRC-20 fungible tokens, ZRC-721 wildlife NFTs with full metadata preservation, ZRC-1155 multi-tokens, and native ZOO/LUX gas tokens. The protocol ensures that conservation metadata, provenance chains, and token-bound account states are faithfully preserved across chains. We describe the bridge gateway contracts, MPC signer set, metadata serialization, TBA freeze/mirror mechanism, fee structure, and liveness/safety guarantees.

Keywords: cross-chain bridge, Warp messaging, MPC, threshold ECDSA, Teleport, metadata preservation

1 Introduction

The Zoo ecosystem exists on Zoo L2 (Chain ID: 200200), a rollup anchored to the Lux primary network. Zoo tokens, wildlife NFTs, and conservation assets must be portable to Lux C-Chain for DeFi liquidity, to Lux X-Chain for exchange listing, and to external chains for broader ecosystem access.

Two bridge architectures serve different trust models. The Lux Warp messaging bridge [1] is the native, trustless path for Zoo L2 $\leftrightarrow$ Lux C-Chain transfers: Warp messages are validated by the Lux primary network validators, requiring no additional trust assumptions. For transfers to external EVM chains, a lock-and-mint bridge secured by an MPC threshold signer set provides Byzantine fault tolerance.

The bridge’s distinguishing feature is conservation metadata preservation. When Tiger #42’s ZRC-721 NFT bridges from Zoo to Lux, its species metadata (IUCN status, GPS coordinates, taxonomy), provenance chain (field observations, veterinary records), and token-bound account state (donation balance, conservation history) must arrive intact on the destination chain.

1.1 Contributions

1. The IZooBridgeGateway interface supporting four asset types (Section 3).
2. MPC threshold signature verification using GG20 (Section 4).
3. Conservation metadata serialization for cross-chain transfer (Section 6).
4. TBA freeze/mirror mechanism for NFT bridge transfers (Section 7).
5. Liveness and safety guarantees with emergency withdrawal fallback (Section 9).
6. Security analysis of MPC key compromise, replay, and censorship attacks (Section 10).

2 Architecture

3 Bridge Gateway

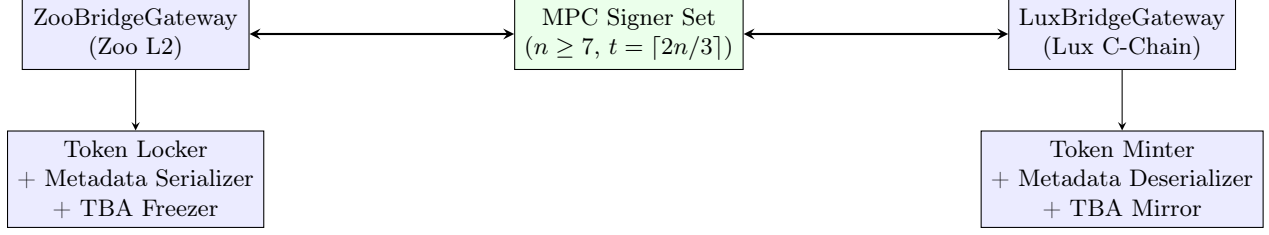


Figure 1: Zoo ↔ Lux bridge architecture.

```

1  interface IZooBridgeGateway {
2      enum AssetType {
3          NATIVE,    // ZOO or LUX gas token
4          ZRC20,     // Fungible token
5          ZRC721,    // Non-fungible token
6          ZRC1155    // Multi-token
7      }
8
9      struct BridgeMessage {
10         bytes32 messageId;
11         uint256 sourceChainId;
12         uint256 destinationChainId;
13         address sender;
14         address recipient;
15         AssetType assetType;
16         address tokenContract;
17         uint256 tokenId;    // 0 for fungibles
18         uint256 amount;    // 1 for NFTs
19         bytes metadata;    // Serialized conservation
20         uint256 nonce;     // Replay protection
21         uint256 deadline;  // Message expiry
22     }
23
24     function bridgeOut(
25         uint256 destinationChainId,
26         address recipient,
27         AssetType assetType,
28         address tokenContract,
29         uint256 tokenId,
30         uint256 amount
31     ) external payable returns (bytes32 messageId);
32
33     function bridgeIn(
34         BridgeMessage calldata message,
35         bytes calldata mpcSignature
36     ) external;
37
38     function mpcSignerSet()
39         external view returns (address);
40     function processedMessages(bytes32 messageId)
41         external view returns (bool);
42
43     event BridgeOutInitiated(
44         bytes32 indexed messageId,
45         address indexed sender,
46         uint256 destinationChainId,

```

```

47     AssetType assetType,
48     address tokenContract,
49     uint256 tokenId,
50     uint256 amount
51 );
52 event BridgeInCompleted(
53     bytes32 indexed messageId,
54     address indexed recipient,
55     uint256 sourceChainId,
56     AssetType assetType,
57     address tokenContract,
58     uint256 tokenId,
59     uint256 amount
60 );
61 }

```

Listing 1: IZooBridgeGateway interface

3.1 Warp Messaging Path

For Zoo L2 ↔ Lux C-Chain transfers, the bridge uses native Warp messaging [1]:

```

1 IWarpMessenger constant WARP =
2     IWarpMessenger(
3         0x020000000000000000000000000000000000000000000000000000000000000005
4     );
5
6 function bridgeOutViaWarp(
7     uint256 amount, address recipient
8 ) external {
9     IERC20(zooToken).transferFrom(
10         msg.sender, address(this), amount);
11     bytes memory payload = abi.encode(
12         recipient, amount);
13     WARP.sendWarpMessage(payload);
14 }
15
16 function bridgeInViaWarp(uint32 index) external {
17     (IWarpMessenger.WarpMessage memory msg,
18      bool valid) =
19         WARP.getVerifiedWarpMessage(index);
20     require(valid, "invalid_warp_message");
21     require(msg.sourceChainID == peerChainId,
22            "wrong_source");
23     (address recipient, uint256 amount) =
24         abi.decode(msg.payload,
25                   (address, uint256));
26     IERC20(zooToken).transfer(recipient, amount);
27 }

```

Listing 2: Warp bridge integration (from ZIP-0022)

Warp messages are validated by the Lux primary network validators. No MPC signer set is needed for this path.

Parameter	Value
Minimum signers (n)	7
Threshold (t)	$\lceil 2n/3 \rceil$
Key generation	Distributed key generation (DKG)
Signature scheme	Threshold ECDSA (secp256k1)
Rotation period	30 days
Slashing	10,000 LUX stake; slashed for equivocation

Table 1: MPC signer set parameters.

4 MPC Threshold Signatures

For external chain bridges, the protocol uses GG20 [3] threshold ECDSA:

```

1  interface IMPCSignerSet {
2      function verifyThresholdSignature(
3          bytes32 messageHash,
4          bytes calldata signature
5      ) external view returns (bool valid);
6
7      function signerSetInfo()
8          external view returns (
9          uint256 totalSigners,
10         uint256 threshold,
11         uint256 rotationEpoch,
12         bytes32 publicKeyHash
13     );
14
15     function proposeRotation(
16         address[] calldata newSigners,
17         uint256 newThreshold
18     ) external;
19
20     event SignerSetRotated(
21         uint256 epoch,
22         uint256 totalSigners,
23         uint256 threshold,
24         bytes32 publicKeyHash
25     );
26 }

```

Listing 3: IMPCSignerSet interface

MPC threshold ECDSA produces a single on-chain signature regardless of signer set size. This costs constant gas for verification, unlike multisig which scales linearly with signer count.

Theorem 4.1 (Byzantine Fault Tolerance). *With $n = 7$ signers and threshold $t = 5$, the bridge tolerates $f = 2$ Byzantine (compromised) signers. An attacker must compromise $t = 5$ signers to forge a bridge message. The economic cost of compromise is $5 \times 10,000 = 50,000$ LUX in slashable stake.*

5 Asset Bridging Mechanics

5.1 Native Token (ZOO/LUX)

- **Zoo to Lux:** ZOO locked in ZooBridgeGateway. Wrapped ZOO (wZOO) minted on Lux as LRC-20 (LP-3800).
- **Lux to Zoo:** wZOO burned on Lux. ZOO unlocked on Zoo L2.
- LUX follows the reverse path (locked on Lux, wrapped on Zoo).

5.2 ZRC-20 Tokens

Tokens locked on the source chain. A bridged representation is minted on the destination per LP-3800. Conservation extensions (donation rate, impact credits) are restored from on-chain state on return.

5.3 ZRC-721 NFTs

1. NFT locked on Zoo L2.
2. Species metadata serialized via ABI encoding and included in `BridgeMessage.metadata`.
3. Provenance chain snapshot uploaded to IPFS; CID included in metadata.
4. TBA frozen (ZIP-0703).
5. Mirror LRC-721 minted on Lux with full metadata.
6. On return: mirror burned, original unlocked, TBA unfrozen, “bridged_back” provenance record added.

5.4 ZRC-1155 Multi-Tokens

Multiple token IDs and amounts can be bridged in a single message. Soulbound tokens MUST NOT be bridged; the gateway rejects any `bridgeOut` for non-transferable token IDs.

6 Metadata Serialization

```
1 bytes memory metadata = abi.encode(  
2     speciesMetadata,           // SpeciesMetadata  
3     provenanceSnapshotCID,    // IPFS CID  
4     conservationFundAddress,  
5     royaltySplit,             // (conserveBps, creatorBps)  
6     tbaAddress,               // TBA on source chain  
7     tbaBalanceSnapshot        // ZOO balance in TBA  
8 );
```

Listing 4: ZRC-721 metadata serialization

ABI encoding ensures deterministic serialization. The destination chain deserializes the metadata and reconstructs the on-chain state. The provenance CID provides a full history reference without including the entire provenance chain in the bridge message.

Algorithm 1 TBA Freeze on Bridge-Out

- 1: **Input:** NFT N , destination chain D
 - 2: Lock N in bridge gateway
 - 3: $\text{TBA.state}() \leftarrow \text{TBA.state}() + 1$ {Invalidate pending ops}
 - 4: $\text{TBA.frozen} \leftarrow \text{true}$
 - 5: Serialize TBA balance snapshot
 - 6: Include in bridge message metadata
 - 7: Relay to destination chain D
 - 8: Create mirror TBA on D with same salt
-

Operation	Fee	Recipient
Native token bridge	0.1%	MPC signers
ZRC-20 bridge	0.05%	50% signers, 50% conservation
ZRC-721 bridge	Flat 10 ZOO	Conservation fund
ZRC-1155 batch	5 ZOO per type	Conservation fund

Table 2: Bridge fee schedule.

7 TBA Freeze/Mirror

The freeze and NFT lock are atomic within a single `bridgeOut` transaction. This prevents a race condition where the TBA is accessed between the lock and the freeze.

Theorem 7.1 (Cross-Chain Asset Safety). *A TBA’s assets cannot be double-spent across chains. The freeze mechanism ensures exactly one chain has an active TBA at any time. Asset access on the inactive chain requires a bridge transfer, which is mediated by the MPC signer set.*

8 Fee Structure

9 Liveness and Safety

Liveness: If the MPC signer set is unavailable for > 72 hours, a governance-controlled emergency withdrawal path activates. Users can reclaim locked assets after a 7-day timelock.

Safety: Messages expire after `deadline` (default 24 hours). Expired messages are rejected. Each `messageId` is processed at most once.

Finality: Bridge messages are relayed only after source chain finality. Zoo L2 inherits sub-second finality from the Lux primary network. Lux C-Chain requires 2 seconds.

Rate limiting: Per ZIP-0022, the bridge enforces per-transaction, hourly, and daily volume limits:

Theorem 9.1 (Supply Invariant). *The bridge enforces:*

$$S_{\text{zoo_l2}} + S_{\text{locked}} = S_{\text{initial}} + S_{\text{emissions}} \quad (1)$$

An on-chain audit function verifies this invariant and pauses the bridge if violated.

Parameter	Value	Adjustable
Per-transaction limit	1,000,000 ZOO	Governance
Hourly volume limit	10,000,000 ZOO	Governance
Daily volume limit	50,000,000 ZOO	Governance
Emergency pause	25% of locked supply in 24h	Fixed

Table 3: Bridge rate limiting parameters.

10 Security Analysis

10.1 MPC Key Compromise

If t signers are compromised, the bridge is vulnerable to fraudulent minting. Mitigations: 30-day key rotation, geographic distribution of signers, HSMs, and slashing for equivocation.

10.2 Replay Attacks

Each `messageId` is tracked in `processedMessages`. The nonce and deadline fields provide additional replay protection.

10.3 Metadata Tampering

The `messageHash` covers all fields including `metadata`. MPC signature verification ensures integrity.

10.4 Censorship

If MPC relayers censor specific messages, the 72-hour liveness fallback allows governance to force-process messages.

10.5 TBA Freeze Race Condition

The TBA must be frozen before the bridge message is relayed. The `bridgeOut` function atomically locks the NFT and freezes the TBA in a single transaction.

10.6 Signer Set Rotation

During rotation, both old and new signer sets are valid for a 24-hour overlap period. Messages signed by either set are accepted during this window.

11 Related Work

Lux Teleport (LP-6332) [2] provides the transport layer; ZIP-0800 adds the application layer for conservation metadata. LayerZero [6] provides generic cross-chain messaging but lacks conservation metadata support. Wormhole [7] uses a guardian set similar to our MPC signer set. Axelar [8] provides general message passing with threshold signatures. Chainlink CCIP [9] offers cross-chain interoperability with oracle-based verification. The Cosmos IBC protocol [10] provides trustless interchain communication for Tendermint-based chains; Lux Warp messaging provides analogous guarantees within the Lux ecosystem.

12 Conclusion

The Zoo $\leftrightarrow$ Lux bridge combines trustless Warp messaging for intra-ecosystem transfers with MPC threshold signatures for external chain access. Conservation metadata preservation ensures that wildlife NFTs, impact credits, and token-bound account states survive cross-chain transfers intact. Rate limiting and the supply invariant provide defense-in-depth against the bridge exploits that have plagued cross-chain DeFi.

Acknowledgments

This work was supported by Zoo Labs Foundation. We thank the Lux team for the Teleport architecture and Warp messaging protocol.

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